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Online prediction of network-level public transport demand based on principle component analysis

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ABSTRACT

Online demand prediction plays an important role in transport network services from operations, controls to management, and information provision. However, the online prediction models are impacted by streaming data quality issues with noise measurements and missing data. To address these, we develop a robust prediction method for online network-level demand prediction in public transport. It consists of a PCA method to extract eigen demand images and an optimization-based pattern recognition model to predict the weights of eigen demand images by making use of the partially observed real-time data up to the prediction time in a day. The prediction model is robust to data quality issues given that the eigen demand images are stable and the predicted weights of them are optimized using the network level data (less impacted by local data quality issues). In the case study, we validate the accuracy and transferability of the model by comparing it with benchmark models and evaluate the robustness in tolerating data quality issues of the proposed model. The experimental results demonstrate that the proposed Pattern Recognition Prediction based on PCA (PRP-PCA) consistently outperforms other benchmark models in accuracy and transferability. Moreover, the model shows high robustness in accommodating data quality issues. For example, the PRP-PCA model is robust to missing data up to 50% regardless of the noise level. We also discuss the hidden patterns behind the network level demand. The visualization analysis shows that eigen demand images are significantly connected to the network structure and station activity variabilities. Though the demand changes dramatically before and after the pandemic, the eigen demand images are consistent over time in Stockholm.

1. Introduction

With the growth of metropolitan regions and the influx of people into urban areas, public transport networks in major cities are experiencing operation and management challenges such as service unreliability and crowding. The mismatch between supply and demand is the main cause of poor service quality and wasted energy in public transport. Passenger demand prediction is important to contribute to effective operations control and management in providing timely information for traffic management and passenger travel suggestions (Gkiotsalitis and Cats, 2021; Halvorsen et al., 2020; Jiao et al., 2021; Tang et al., 2018). The Coronavirus pandemic (Covid-19) crisis has greatly impacted public transport ridership and service provision across the world, and for example, ridership has dropped around 60% in Stockholm in 2020, and detailed spatiotemporal analysis of the ridership changes could refer to Jenelius and Cebecauer (2020). As many countries start to navigate their return to normality, new public transport planning requirements are devised. These measures imply a major reduction in service capacity compared to the era before Covid-19. At the time of writing, there was a severe lack of knowledge regarding the potential impact of the pandemic on public transport operations and models that can support transport service planning under these new challenges. We systematically reviewed and synthesized the literature on the impacts of COVID on public transport to identify the necessity of adjusting planning measures. On the other hand, the existing methods for public transport planning are at the strategic, tactical, and operational levels. We identify intervention measures that can support public transport service providers in planning their services in the post-shutdown phase and their respective modeling development requirements. This can support the transition from the initial ad-hoc planning practices to more evidence-based decisionmaking. The ubiquitous automated data collection techniques in public transport systems lend the opportunity for real-time demand prediction in practical applications (Koutsopoulos et al., 2019; Pelletier et al., 2011).

The passenger demand prediction has attracted the most attentions in

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literature, either for entry or exit demand (Jiang et al., 2022). The short-term (online) passenger demand prediction is important to contribute to public transport management, which helps to provide timely information, implement responsive measures, and improve the level of service. (Karlaftis and Vlahogianni, 2011; Tedjopurnomo et al., 2020; Yin et al., 2022; Wei and Chen, 2012). For example, short-term predictions can reveal the crowded stations in the transport system and the online schedule and passenger management can be conducted to relieve the situation. Generally, the prediction models can be classified into statistical, machine learning, deep learning, and hybrid models. For statistical models, commonly used models include Auto-Regressive (AR), Moving Average (MA), AutoRegressive Integrated Moving Average (ARIMA), and historical average models. The statistical models are easy to build and have high interpretability; however, they were limited capability in capturing complex and nonlinear relationships between variables. Machine learning models are reported to have better prediction performance given their model-free characteristics and capabilities in modeling complex nonlinearities. The commonly used models include support vector regression, random forest regression, and regression trees (Li et al., 2019; Xia et al., 2013; Xu et al., 2020). The deep learning models are capable of modeling highly nonlinear relationships in a multivariate setting and many studies reported that they perform better than typical machine learning models if fully trained (Kosolsombat and Saraubon, 2018). The typical models used are Long Short Term Memory (LSTM), Recurrent Neural Networks (RNN), and Convolutional Neural Networks (CNN) (Jiang et al., 2022; Oh et al., 2015). Hybrid models consist of a mixture of different models, which involve the combination of two or more base algorithms to build a more robust standalone algorithm. The interest in hybrid models has become more avid in recent years due to their ability to make the best use of the complementary performance of original methods (Büyükşahin and Ertekin, 2019).

Basically, the above prediction studies belong to the time-series prediction problem (using historical or real-time time series for future predictions). Although they provide good predictions in the tested dataset, these models are developed with the assumption that the input time series data is accurate which is not realistic for practical deployment. In practice, the data quality issues are significant while using Automatic Fare Collection Data (AFC data) due to infrastructure failure and data management problems (Zhang et al., 2022). For example, more than 60% of stops/stations cannot be recognized in AFC dataset due to the issues with General Transit Feed Specification (GTFS) data in South Queensland (Nassir et al., 2015). However, limited studies are found in the literature that model online prediction problems considering data quality issues for practical deployments.

Generally, there are two approaches to addressing data quality issues in online predictions from either the perspective of the input data (e.g., data cleaning or missing data imputation before using them for prediction) or the prediction methodology (developing prediction models acknowledging data quality issues). The first approach is to improve data quality before prediction (Arentze et al., 2000; Turner, 2004). For example, Korn et al. (2003) applied a monitoring method using probabilistic, approximate constraints (PACs) to get reliable data in an aggregated network. Turner et al. (2000) analyzed intelligent transport system data by evaluating the erroneousness, completeness, and accuracy to make an error detection rule to screen for erroneous or suspect data. However, the data pre-processing procedure is complicated and time-consuming, especially for large volumes of streaming data.

This paper focuses on developing robust online network-level demand prediction methods considering data quality issues in practice. The general idea for such an approach is to develop models using inputs of hidden demand patterns as opposed to raw observations. Several studies developed prediction models based on extracted demand patterns using pattern recognition methods, such as principal component analysis (PCA). For example, Luo et al. (2017) used PCA to multivariate time periods and flows throughout a month to reveal approximatively the passenger flows based on metro smart card data in Shenzhen. Jiang et al. (2020) applied the PCA method in Hong Kong to identify unobserved activities based on various eigen demand images and predict MTR passenger demand. Xing et al. (2015) introduced an approach to predict real-life data by using the robust PCA with a better decomposition performance and more accurate spatial traffic flow pattern, which can lead to a more reliable analysis and prediction. The PCA can extract the hidden principal components of demand patterns which represents the observed demand time in space and time using low-dimension matrices. It captures the 'hidden' fundamental mechanisms (e.g., travel activities and choices) governing the 'observed' data generation process (i.e., demand and flows), thus they are robust to observe noises. The hypothesis is that we only need a small number of eigen demand images in order to reconstruct any demand matrix at any time (e.g., network-level demand matrix at 7:00–8:00) observed in practice. The hypothesis is conceptually valid for public transport travels since human travel behavior has a strong regularity (Prabhu Teja and Ravi, 2012). We define the eigen demand image as the extracted hidden demand patterns representing station demand organized geographically in a network. The eigen demand images capture the 'hidden' demand generation mechanisms which have a high potential for robust demand prediction applications with data quality issues (Gan et al., 2020; Lopez et al., 2017).

The PCA based prediction studies provide a good reference for our study. However, they focus more on reconstructing the time-interval demand for a whole day in the future rather than predicting the real-time network-level demand in future time intervals within a day using the partially observed real-time demand up to the time when the prediction is made. The paper proposes an eigen demand classification based model for robust online network-level public transport demand prediction using the partially observed real-time demand data. The model has two main steps: 1) generating the eigen demand images from historical real-time demand data using the PCA model; 2) predicting the network-level demand in future time intervals using real-time demand observations by solving a pattern recognition optimization problem (more details in the Methodology section). The main contributions of the paper are:

- Propose a robust online network-level demand prediction approach using the PCA method and an optimization based pattern recognition model in public transport. It accommodates data quality issues for online predictions and makes the best use of partially observed real-time data for predictions within a day;
- Systematically validate the model performance using public transport demand data in Stockholm in terms of accuracy, robustness, and transferability;
- Discuss the interpretation of generated eigen demand images in practice, and explore their differences before and after Covid-19.

The remaining paper is organized into the following sections: The methodology section defines the online prediction problem with data quality issues. It proposes the method for generating eigen demand images and the pattern recognition model for online demand prediction. The case study comprehensively illustrates the interpretation of eigen demand images in practice and validates the proposed model (accuracy, robustness, and transferability) by comparing it with baseline models using public transport data in Stockholm. The final section summarizes the main findings and future research directions.

2. Methodology

2.1. Problem description

Generally, the online passenger demand prediction is modeled as a time series prediction problem. Mathematically, the future demand is predicted as a function of historical time series capturing timely, daily, weekly temporal patterns and spatial correlations (Halyal et al., 2022; Yang et al., 2019).

However, these models do not consider the online data quality issues

for practical deployment. They assume that the input data are accurate for testing which is unrealistic with large-volume streaming data collected from the AFC system in practice. As discussed in the introduction, there are two general approaches to address data quality issues: 1) improving data quality before predictions, or 2) developing robust prediction models. We focus on the second approach and develop a robust prediction method for online network-level demand prediction in public transport using the PCA method and an optimization-based pattern recognition model. The eigen demand images extracted by PCA reflect the underlying network structure, operations, and demand patterns, which are stable to events such as Covid. We assume the training data are ground truth since historical data can be checked and preprocessed, and then these images can be seen as "true parameters" for a specific area in the prediction. With the "true parameters", the influences by noises in partial information (observations) used for prediction are limited. Besides, the online demand prediction also improves the model robustness to the tolerance of data quality issues because the Ordinary Least Square, recognizing the daily patterns, adjusts the deviation of erroneous observations for the predictions.

Fig. 1 shows the framework of the proposed method. It includes two main modules: eigen demand image generation and online demand prediction. Note that the demand reconstruction module is provided to show the difference between the studies we forced and the literature.

- Eigen demand image generation: It extracts hidden eigen demand images and keeps important ones for further use. It first generates the spatiotemporal demand image from the input demand data, then extracts eigen demand images using the PCA method;
- Online demand prediction. It predicts the network-level demand at future time intervals within a day using partially observed real-time demand data. An optimization model is developed to estimate the weights of eigen demand images given real-time observations used to predict the network-level demand of next time.

2.2. Eigen demand image generation

The network-level public transport demand can be represented as an image with stations/grid cells as pixels. Similar to the eigenfaces in image processing, eigen demand images could be extracted using demand images over different times and days. We construct a space demand image in a certain time interval (e.g., 10:00–10:30) with a size $M*N$ using a geographical grid system, where $M*N$ is the number of valid grids (with at least one station in each grid) partitioned in an area with a certain grid size.

$$\gamma_{kq} = \begin{bmatrix} x_{11} \\ x_{12} \\ \vdots \\ x_{M1} \\ \vdots \\ x_{MN} \end{bmatrix} \tag{1}$$

where γ_{kq} is space demand image at time interval k of day q and x_{mn} is the demand of the grid cell indexed by (m, n) in the grid-based map.

Suppose there are K time intervals (e.g., 30 min) in a day and Q observation days in the dataset. The spatiotemporal demand image $\boldsymbol{D} \in \mathbb{R}^{(K*M*N)\times Q}$ is expressed as

$$\boldsymbol{D} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & & \gamma_{1Q} \\ \gamma_{21} & \gamma_{22} & \cdots & \gamma_{2Q} \\ \gamma_{31} & \gamma_{32} & & \gamma_{3Q} \\ \vdots & & \ddots & \vdots \\ \gamma_{K1} & \gamma_{K2} & \cdots & \gamma_{KQ} \end{bmatrix} \tag{2}$$

For each column in the matrix $\boldsymbol{D}$, it aggregates the space demand images of all time intervals of a certain day. For each row in the matrix $\boldsymbol{D}$,

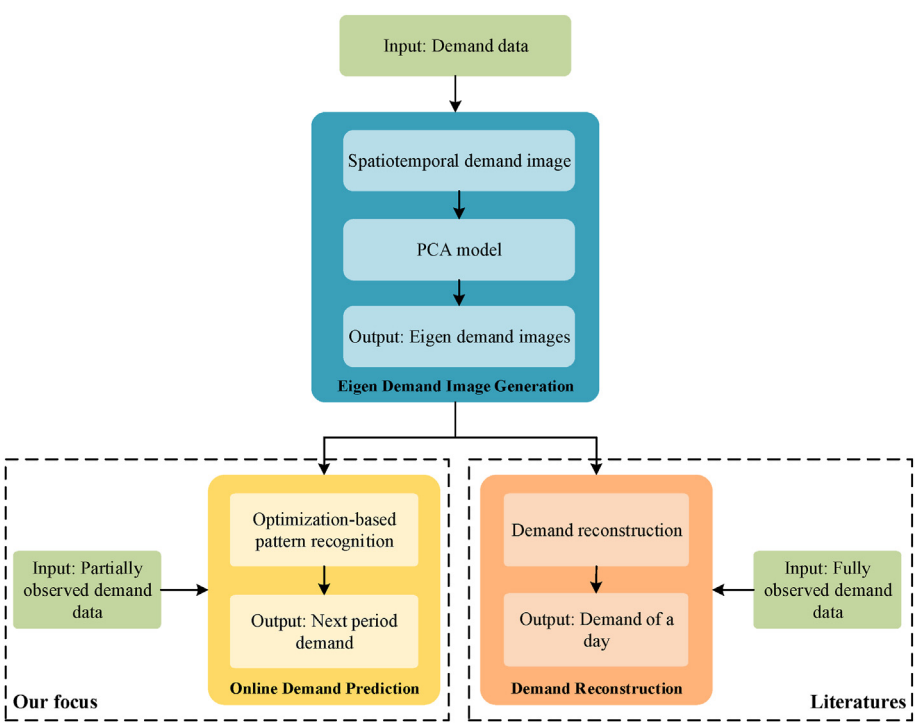

Fig. 1. Methodology framework for the online demand prediction (our focus) and reconstruction (in literature).

it represents the space demand images of different days at the same time interval.

Therefore, the average demand $\overline{\boldsymbol{D}}$ for a day is calculated:

$$\overline{\boldsymbol{D}} = \frac{1}{Q}\sum_{q=1}^{Q}\begin{bmatrix}\gamma_{1q}\\ \gamma_{2q}\\ \gamma_{3q}\\ \vdots\\ \gamma_{Kq}\end{bmatrix} \tag{3}$$

where the $[\gamma_{1q}, \gamma_{2q}, \ldots, \gamma_{Kq}]^{\mathrm{T}}$ is the space demand image of day q.

The residual demand of day q can be obtained by subtracting the average demand $\overline{\boldsymbol{D}}$ from the space demand image of day q.

$$\boldsymbol{\Phi}_q = \begin{bmatrix}\gamma_{1q}\\ \gamma_{2q}\\ \gamma_{3q}\\ \vdots\\ \gamma_{Kq}\end{bmatrix} - \overline{\boldsymbol{D}} \tag{4}$$

Similar with the work (Jiang et al., 2020), the covariance matrix $\boldsymbol{C}$ of the daily residual demand across different days is:

$$\boldsymbol{C} = \frac{1}{Q}\sum_{q=1}^{Q}\boldsymbol{\Phi}_q\boldsymbol{\Phi}_q^{\mathrm{T}} = \boldsymbol{A}\boldsymbol{A}^{\mathrm{T}} \tag{5}$$

where matrix $\boldsymbol{A} = [\boldsymbol{\Phi}_1, \boldsymbol{\Phi}_2, \ldots, \boldsymbol{\Phi}_Q] \in \mathbb{R}^{(K*M*N)\times Q}$.

Then we can calculate the ith eigenvalue λ_i and the corresponding eigenvector $\boldsymbol{u}_i$.

$$\boldsymbol{C}\boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i \tag{6}$$

To reduce the computational complexity, we set matrix $\boldsymbol{L} = (\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})$ and calculate the jth eigenvalue β_j and the corresponding eigenvector $\boldsymbol{v}_j$. This is a common step for PCA method in the face recognition to reduce computational complexity. It preserves the information of the original covariance matrix and improves the efficiency of the PCA, as the original matrix $\boldsymbol{C}$ with the size of $(K*M*N)*(K*M*N)$ is not used in the iteration but the smaller matrix $\boldsymbol{L}$ with the size of $(Q*Q)$ (Chaudhari, 2012). Eq. (6) can be transformed into:

$$\boldsymbol{L}\boldsymbol{v}_j = \beta_j\boldsymbol{v}_j\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}v_j = \beta_j\boldsymbol{v}_j\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}v_j = \boldsymbol{A}\beta_j\boldsymbol{v}_j\boldsymbol{C}(\boldsymbol{A}\boldsymbol{v}_j) = \beta_j(\boldsymbol{A}\boldsymbol{v}_j) \tag{7}$$

where $\boldsymbol{v}_j$ is the jth eigenvalue of $\boldsymbol{L}$ and β_j is the corresponding eigenvalue.

For the covariance matrix $\boldsymbol{C}$, its eigenvector $\boldsymbol{u}_j = \boldsymbol{A}\boldsymbol{v}_j$. The eigenvectors of the covariance matrix $\boldsymbol{C}$ are used as eigen demand images.

2.3. Online demand prediction

The pattern recognition based online demand prediction aims to identify the demand pattern of a day based on the observed demand data to predict the demand of the next time period. Inspired by facial recognition, any daily demand image can be represented by a linear combination of eigen demand images and projections (weights) (Turk and Pentland, 1991). The projections, also called weights, show how much a certain pattern presents in a demand image of a specific day. The weight vector is presented as $\boldsymbol{W} = [w^1, w^2, w^3, \ldots, w^J] \in \mathbb{R}^{J\times 1}$, where w^j is the weight of jth eigen demand image and J is the kept number of eigen demand images. In the prediction, the eigen demand images are extracted in Section 2.2 and the space demand image $\boldsymbol{D}_{1-k}$ is observed up to the time interval k when the prediction is made. The objective is to estimate the weight vector $\widehat{\boldsymbol{W}}_k$ (estimated eigen demand image weights) using the observed demand information up to the prediction time k and to predict the network-level demand in the next time interval. Formally, the online demand prediction problem is:

Given the space demand image $\boldsymbol{D}_{1-k}$ of a certain day, space average demand $\overline{\boldsymbol{D}}_{1-k}$, and partial eigen demand image $\boldsymbol{V}_{1-k}$ up to the prediction time interval k, to estimate the weight vector $\widehat{\boldsymbol{W}}_k$ and predict the network-level demand $\boldsymbol{D}_{k+1}$ of the next time interval $(k+1)$.

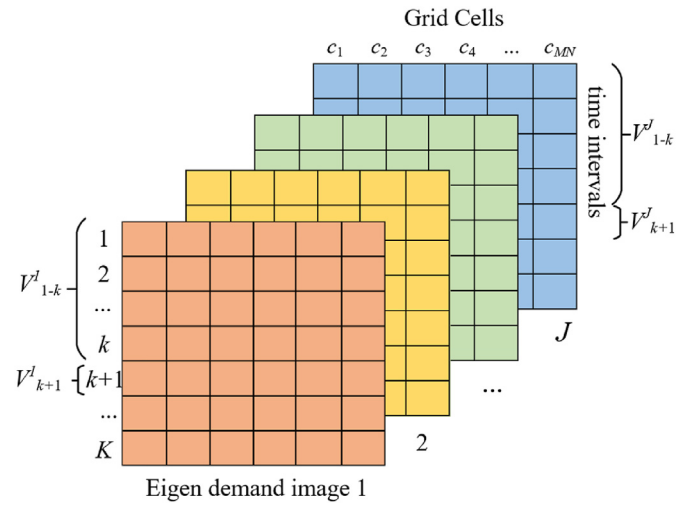

Fig. 2. Illustration of partial observed and used eigen demand images for online demand prediction.

Fig. 2 illustrates the full eigen demand images with the row representing time intervals of a day and the column for stations. It also shows the partial j^{th} eigen demand image $\boldsymbol{V}^j_{1-k}$ observed from time 1 to k, and $\boldsymbol{V}^j_{k+1}$ used for demand prediction at the next time interval $k+1$.

The problem of calculating the weight vector $\widehat{\boldsymbol{W}}_k$ at time interval k is an overdetermined system ($\boldsymbol{D}_{1-k} = \sum_1^J \boldsymbol{V}^j_{1-k}\widehat{w}^j_k$) because there are more equations than unknown parameters. The optimization objective to minimize the sum of squares of the differences between observed and predicted demand, which can be achieved by Ordinary Least Square (Pavic and Novoselac, 2017; Williams, 1990), is shown in Eq. (8):

$$\min_{\widehat{\boldsymbol{W}}_k} \sum_1^J \boldsymbol{V}^j_{1-k}\widehat{w}^j_k - (\boldsymbol{D}_{1-k} - \overline{\boldsymbol{D}}_{1-k}) \tag{8}$$

where matrix $\boldsymbol{V}^j_{1-k} \in \mathbb{R}^{k\times(M*N)}$ is the jth eigen demand image up to time interval k, value $\widehat{w}^j_k$ is the weight of $\boldsymbol{V}^j_{1-k}$ to be estimated, and $\boldsymbol{D}_{1-k} \in \mathbb{R}^{k\times(M*N)}$ is the partial observed space demand image up to time interval k, and $\overline{\boldsymbol{D}}_{1-k} \in \mathbb{R}^{k\times(M*N)}$ is the partial average demand matrix up to the time interval k.

Hence, the network-level demand for the next time interval $\widehat{\boldsymbol{D}}_{k+1}$ is predicted as

$$\widehat{\boldsymbol{D}}_{k+1} = \sum_1^J \boldsymbol{V}^j_{k+1}\widehat{w}^j_k + \overline{\boldsymbol{D}}_{k+1} \tag{9}$$

where vector $\boldsymbol{V}^j_{k+1} \in \mathbb{R}^{1\times(M*N)}$ is the jth eigen demand image of the next time period $k+1$, vector $\overline{\boldsymbol{D}}_{k+1} \in \mathbb{R}^{1\times(M*N)}$, the corresponding historical network-level average demand. The frequency of the prediction is flexible to set based on needs (e.g., 30 min or 1 h).

3. Case study

The case study uses the public transport 30-min aggregated ticket validation data in Stockholm, Sweden. The smart card system collects every passenger's tap-in data when entering a station/stop. In this study, the station-based dataset includes 1,356,710 records from 05:00 to 22:00 between 1/7/2019–30/11/2019 and 782,756 records between 1/9/2021–30/11/2021 for all public transport modes (buses, trams, and metro). The network (2939 km^2) is divided into 1376 grid cells with a cell size of 1.5 km × 1.5 km, among which 700 and 661 valid grids in 2019

and in 2021 respectively, with at least one station in each cell, are used for further experiments. The prepared data records contain grid cell ID, date, time period, and accumulated passenger counts. We used the elbow method to choose the number of eigen demand images based on their explained variabilities shown in Fig. A1 in Appendix.

3.1. Interpretation of eigen demand images

To explore the interpretation of the eigen demand images, we map the eigen demand images generated from the dataset of 2019 to the geographic location, as shown in Fig. 3. These four eigen demand images are selected and J is the kept number of eigen demand images based on their eigen values. The eigen demand image cell values are colored with the color range from −0.587 to 0.513, where the signs represent demand correlations between grid cells and the magnitudes represent the variability of a grid (Jing et al., 2006). By comparing with the distribution of the physical public transport network, we can find certain interpretations, though more rigorous approach should be designed to understand their physical meanings. Notably, the Eigen image 0 captures the demand patterns around major stations located in central cities. In Eigen image 1, the blue grids correspond to the metro network within the Stockholm city area. Eigen image 2 shows scattered yellow colored grids and varied colors for stations in the central city area. In Eigen image 3, the yellow grids cover almost all stations, probably capturing the average level of variability of demand at stations, while the orange grids to the rail networks in Stockholm. The varied colors for stations in the CBD area appear in all Eigen images, which highlights the relative higher variability of demand in these areas.

Many papers reported that demand patterns change before and after COVID in most cities, which is as expected at the absolute demand level (Beck et al., 2021; Jenelius and Cebecauer, 2020; Sui et al., 2020). The question is whether eigen demand images of the city also change due to COVID. The hypothesis is that the eigen demand image reflects the hidden mechanism of demand generations impacted by three categories of factors, including the city public transport network, operations scheduling, and passenger travel patterns. Fig. 4 shows the eigen demand images generated from the dataset of 2021 with the same scale and colors as that in 2019. It can be observed that the eigen images in 2021 are very similar to those in 2019 (Fig. 3). The difference in Eigen images between 2019 (Fig. 3) and 2021 (Fig. 4) was insignificant ($t(5,190) = -1.21e-15$; $p = 0.99$). It indicates that the pandemic seems to have marginal influences on the hidden pattern of the public transport spatial-temporal demand distribution in Stockholm. We can conclude that even though the ridership decreased after Covid, the eigen demand images are still the same. Therefore, the eigen demand images generated from data 2019 (before Covid) can also be used for the prediction in 2021 (after Covid), which explained the good performance of our model in transferability.

3.2. Online demand prediction

This section illustrates the models, experiment settings, and measurement metrics. The following prediction models are developed for the comparative analysis.

- History Average (HA) Method: Using the historical average demand as the future demand. For example, the next Monday demand is the mean of all historical Mondays demand.

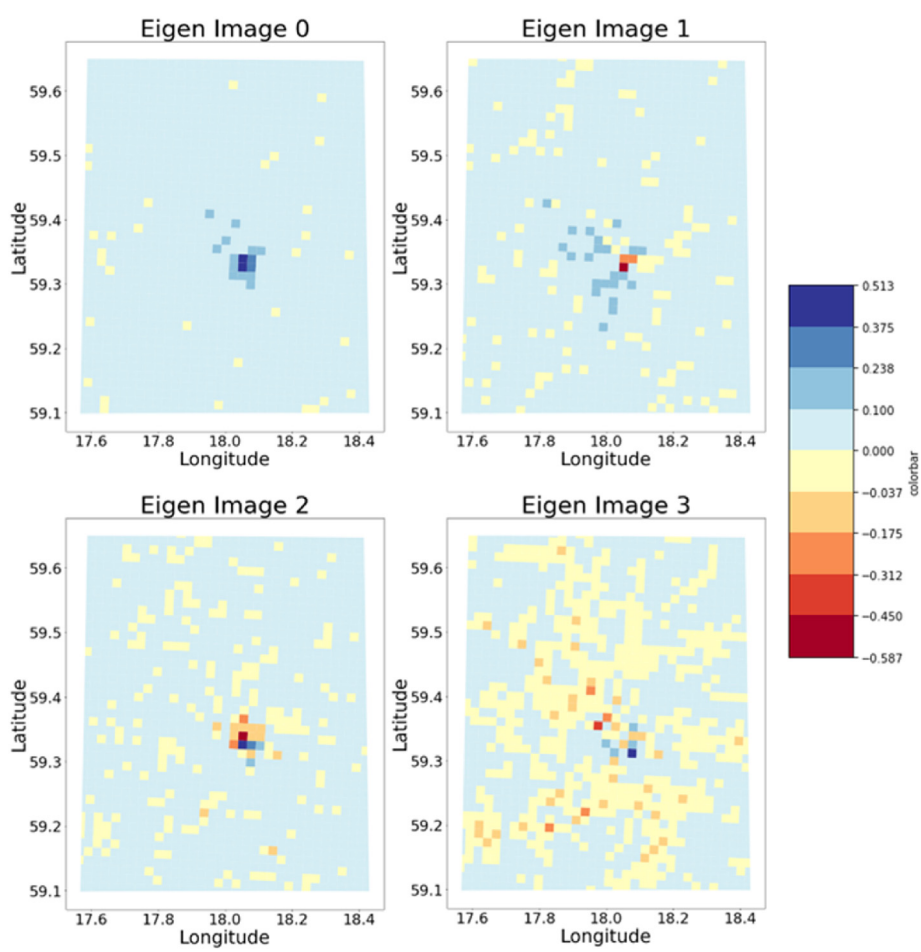

Fig. 3. Eigen demand images of 2019 (1/7/2019–30/11/2019).

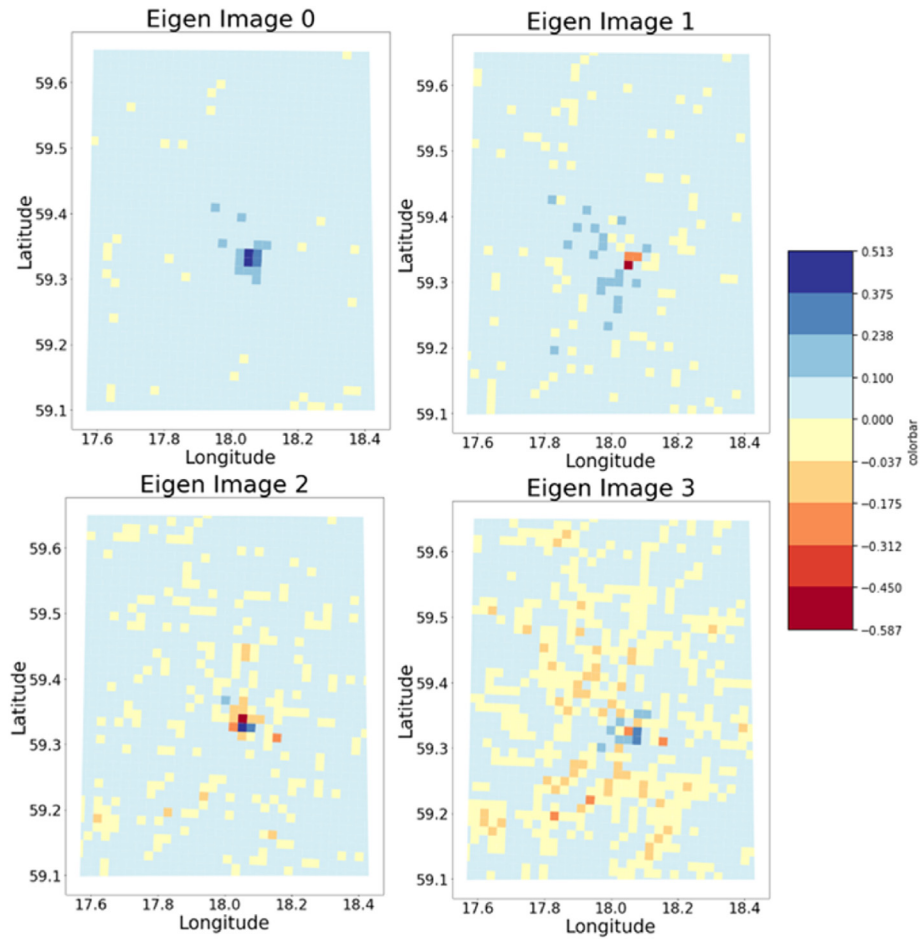

Fig. 4. Eigen demand images of 2021 (1/9/2021–30/11/2021).

- Clustering-PCA(C-PCA) Method: Applying k-means on decomposed demand matrix, assigning day features to each cluster, and using the average reconstructed demand of a cluster as the predicted demand of a corresponding day with the same day feature (Jiang et al., 2020). The eigen demand images are extracted from dataset in 2019.
- Random Forest Regression (RFR): Making the prediction based on weekly information, timely information, and one-interval online observations. For example, the prediction of 06:00–06:30 of a Monday is made by the observations of 05:30–06:00 on the same day, the sequence of intervals in the day (3rd interval in the day), and Monday (denoted as 1).

Table 1
Summary of experiment settings.

Performance	Eigen demand images	Prediction period	Dataset type	Benchmark models	Metrics
Accuracy	5 images extracted from dataset 2019	Nov. 2019	Normal dataset	C-PCA, HA, PRP-PCA1, PRP-PCA4, PRP-PCA8, RFR, LSTM	RMSE, MAE
Transferability		Nov. 2021	After Covid dataset	C-PCA, HA, PRP-PCA1, PRP-PCA4, PRP-PCA8	RMSE, MAE
Robustness		Nov. 2019	Dataset with noises	PRP-PCA1, RFR	RMSE

- Long-short term memory (LSTM): Using 18 intervals of online observations to predict one-interval demand.
- Three pattern recognition predictions based on PCA (PRP-PCA) models proposed with different prediction intervals of 30 min (PRP-PCA1), 2 h (PRP-PCA4), and 4 h (PRP-PCA8)

The metrics of Root Mean Squared Errors (RMSE) and Mean Absolute Errors (MAE) are used to assess the model performance.

$$\mathrm{RMSE} = \sqrt{\frac{1}{M \times N} \sum_{m=1}^{M} \sum\nolimits_{n=1}^{N} (x_{mn} - \widehat{x}_{mn})^2} \tag{10}$$

$$\mathrm{MAE} = \frac{\sum_{m=1}^{M} \sum_{n=1}^{N} |x_{mn} - \widehat{x}_{mn}|}{M \times N} \tag{11}$$

where x_{mn} and $\widehat{x}_{mn}$ are the real observed and predicted demand in indexed by (m, n) for one time period. $M \times N$ is the total number of grid cells in the network.

Table 1 shows the designed experiments to evaluate the model performance in terms of accuracy, transferability, and robustness. The eigen demand images are all extracted using the dataset in 2019. For the accuracy assessment, all benchmark models are tested on the Nov 2019 data. To test the transferability, the trained models in 2019 before Covid are applied to predict post-Covid demand in Nov 2021 to evaluate if the proposed method could be generalized across years, specifically under the pandemic. In terms of robustness, we randomly added noises in the input demand data under certain indicators: missing rate (the percentage

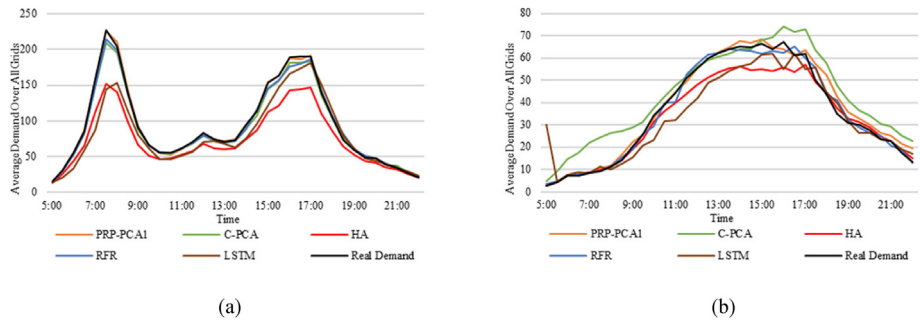

Fig. 5. Predicted demand by different models and the observed demand on (a) Tuesday, Nov 5th in 2019 and (b) Sunday, Nov 10th in 2019.

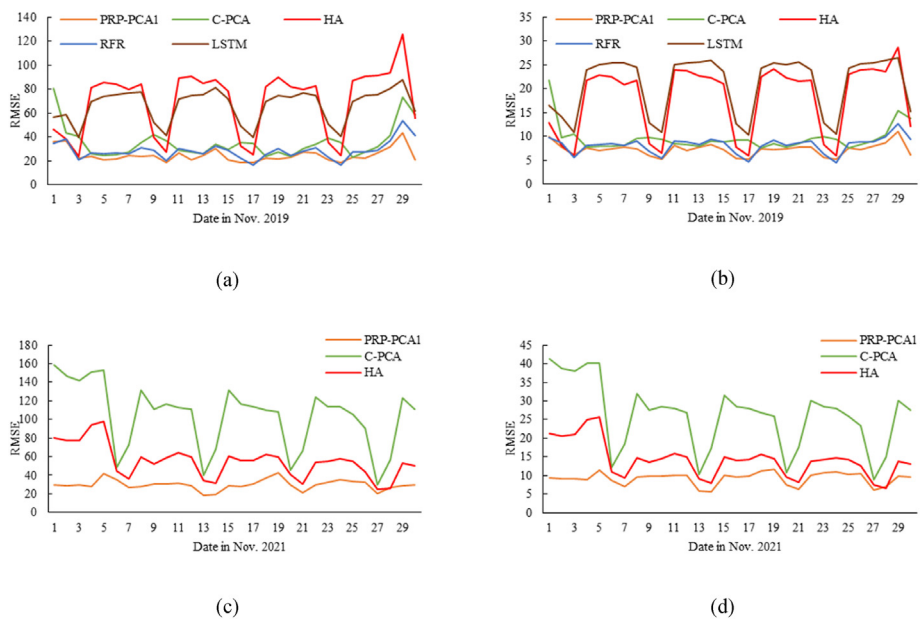

Fig. 6. Prediction performance of PRP-PCA1, C-PCA, HA, RFR and LSTM in terms of accuracy: (a) RMSE, 2019, (b) MAE, 2019; and transferability: (c) RMSE, 2021 and (d) MAE, 2021.

Table 2
Average accuracy and transferability of baselines over 30 days.

Model	Accuracy (2019)		Transferability (2021)	
	RMSE	MAE	RMSE	MAE
PRP-PCA1	24.62	7.38	29.84	9.18
PRP-PCA4	27.70	7.86	34.25	10.16
PRP-PCA8	30.51	7.84	38.21	10.14
C-PCA	35.47	9.64	104.04	26.18
HA	69.96	18.04	55.07	14.21
RFR	28.46	8.21	—	—
LSTM	66.51	20.94	—	—

of grid cells with empty values) and noise level (the percentage errors to real observations in grids).

3.3. Prediction performance comparisons

Fig. 5 shows the comparison between predicted and observed demand for a weekday (Fig. 5(a)) and weekend (Fig. 5(b)). The results show that the PRP-PCA1 model consistently performs better than other models regardless of prediction intervals and has a close prediction to the observations. The second best is the RFR model, whose predictions are closely next to the real demand. The HA model and LSTM model underestimate the demand on both weekday and weekend days, especially during peak hours. The C-PCA model performs better than the HA model and LSTM model on the weekday, but it overestimates the Sunday demand during morning and evening peak hours. It performs worse than the proposed PRP-PCA1 method since it does not take into consideration of the real-time observations for the specific day for the prediction.

Fig. 6 illustrates the performance comparison of baselines on accuracy and transferability based on RMSE and MAE. Generally, the PRP-PCA1 model, based on the eigen demand image generated from the dataset in 2019, performs better compared with other models in terms of accuracy and transferability over 30 days. In terms of accuracy, the PRP-PCA1 model has the lowest and most stable RMSE and MAE over the period. It performs slightly better than the RFR and C-PCA models in terms of RMSE and MAE. LSTM and HA have the worst performance in predicting future demand at the network level. For transferability performance (Fig. 6(c) and 6(d)), we compare three models and the machine learning models (RFR and LSTM) are not able to do the prediction because the inputs/outputs (valid grids or stations) are different for 2019 and 2021. The PRP-PCA1 also performs consistently better than the C-PCA and HA models. Interestingly, the C-PCA model has the largest errors and significant fluctuations with a maximum RMSE of 160 passengers per 30 min, which could be caused by the changes in hidden travel patterns before and after Covid.

Table 2 summarizes the average performance of baselines on accuracy and transferability over 30 days. The PRP-PCA1 has the best performance both on accuracy (RMSE of 24.62 and MAE of 7.38) and

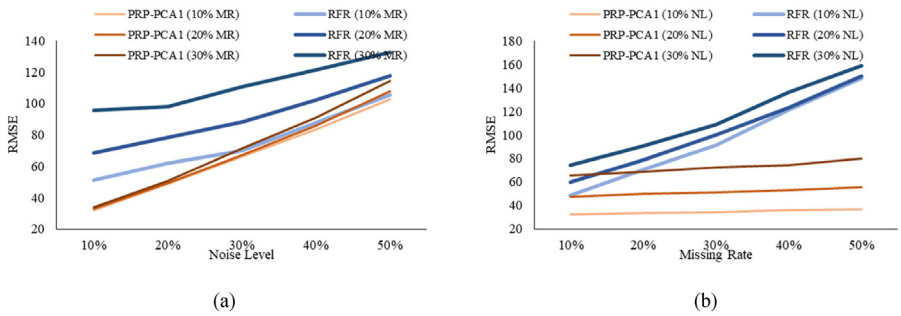

Fig. 7. Performance of PRP-PCA1 with different levels of the (a) Noise Level (NL), and (b) Missing Rate of Observations (MR).

Table 3
Comparisons of model performance under different grid sizes.

Model	Accuracy (1.5 km × 1.5 km)		Accuracy (1 km × 1 km)	
	RMSE	MAE	RMSE	MAE
PRP-PCA1	24.62	7.38	17.36	5.14
PRP-PCA4	27.70	7.86	19.34	5.48
PRP-PCA8	30.51	7.84	20.62	5.31
C-PCA	35.47	9.64	26.91	6.93
HA	69.96	18.04	46.32	11.56
RFR	28.46	8.21	20.90	5.89
LSTM	66.51	20.94	47.80	14.42

transferability (RMSE of 29.84 and MAE of 9.18). With the increase of the prediction time interval h, corresponding to the models of PRP-PCA1, PRP-PCA4, and PRP-PCA8, RMSE shows an increasing pattern while the MAE fluctuated slightly. Except for the proposed model, the RFR model performed the best with slightly worse performance than the PRP-PCA4 model, but the RFR model and LSTM models cannot transfer to predict the data in 2021 because the inputs/outputs (valid grids/stations) are different for 2019 and 2021. The LSTM model and HA model show similar accuracy but the prediction of LSTM has fewer outliers because the RMSE is smaller. The C-PCA model has an average accuracy of 35.47 (RMSE) and 9.64 (MAE) which are almost half of those of HA, but HA model has much fewer errors predicting the demand in the year 2021 (measuring transferability).

To explore the robustness of the proposed model, we compare the performance of the optimal model PRP-PCA1 and the second best model RFR which is proved to be robust to data quality issues (Folleco et al., 2008; Hamza and Larocque, 2005). Fig. 7 shows the performance comparison under different noise levels and missing ratios of observations. As expected, the increase in noise and missing data will decrease the predicted performance for both models. For example, the PRP-PCA model is not sensitive to missing data ratios even 50% missed no matter how much the noise level is. (Fig. 7(b)). Comparatively, the RFR model is relatively more sensitive to missing ratio than the noise level. For example, a 10% increase in the missing rate will lead 20 RMSE increase. Fig. 7(a) shows that the performances of both models are not robust to measurement noises but the performance of the PRP-PCA model is acceptable when the noise level is low. To be specific, with the measurement noise level at 10%, the model prediction RMSE increases around 5 passengers per 30 min per cell compared to no measurement noise (Table 2) regardless of the missing data ratios. In summary, the PRP-PCA1 model shows high robustness in ignoring missing data and tolerating measurement noise under a certain level.

Table 3 shows our model and benchmark models with different cell sizes of 1.5 km × 1.5 km and 1 km × 1 km. The prediction accuracy increases with the decrease of the cell size, which is consistent with (Duan et al., 2019; Ma et al., 2021). The reductions of RMSEs of HA and LSTM are much larger than these of other models. For the percentage of change, it is interesting that the RMSEs and MAEs for all models are reduced by about 1/3 which is consistent with the 1/3 reduction of the length of grid edges. In practice, setting the cell size is tricky which is constrained by downstream applications and the prediction accuracy itself. The test results above show that the proposed model consistently performs better than other benchmark models regardless of the size of the grids.

Note that, most models conduct time series predictions based on observed patterns, and our proposed model also considers hidden patterns of observations. As shown in Figs. 3 and 4, the eigen demand images (hidden patterns) are not changed after and before Covid, which further proves the transferability of our model. This finding can, on the other hand, support the robustness of the proposed model to data quality issues. In our experiments, the PRP-PCA models outperform other benchmark models in terms of accuracy, transferability, and robustness.

4. Conclusions

This paper introduces a new online prediction problem with data quality issues which is critical for practical deployments but rarely studied in the literature. We develop a robust prediction method for online network-level demand prediction in public transport using the PCA method and an optimization-based pattern recognition model. It accommodates data quality issues for online predictions and makes the best use of the partially observed real-time data for predictions during a day.

We validate the model performance in terms of accuracy, transferability, and robustness using the ticket validation data in Stockholm public transport in 2019 and 2021 and compare it with benchmark models. The results show that the proposed PRP-PCA model consistently outperforms other benchmark models in all studied aspects. In addition, the model shows high robustness particularly in missing data and measurement noises. For example, the PRP-PCA model is robust to a missing data ratio up to 50%no matter how much the measurement noise level is. With the measurement noise level at 10%, the model prediction RMSE only increases around 5 passengers per 30 min per cell compared to the case no measurement noise (Table 2) regardless of the missing data ratio. The interpretation results shows that eigen demand images are highly corresponding to the network structure and activity variability levels. It also shows that the eigen demand images are stable over time, even before and after COVID under which the demand using PT was dramatically reduced.

The study has important implications for practice. One critical aspect is to consider data quality issues when modeling the online prediction problem in transportation applications. The pattern recognition-based robust prediction approach shows a good potential for accommodating data quality issues in public transport, which could be extended to other predictions, such as traffic flows and travel times. Future work could develop advanced models for eigen demand extraction (e.g., Robust PCA) and online demand prediction (e.g., deep learning models) to improve the prediction performance. Also, as common for the learning-based approach, physically interpreting eigen demand images is very

challenging, and it would be interesting to develop methodologies to interpret their physical meaning (e.g., explainable AI algorithms).

Replication and data sharing

The author has uploaded the code to an explicit Github repository at. https://github.com/Frankkeepgoingyeah/CMMTR-project.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Fig. A1 shows that the marginal increases in the percentage of variance become insignificant after 4 eigen demand images. Thus, the 4 images are chosen for analysis in the paper

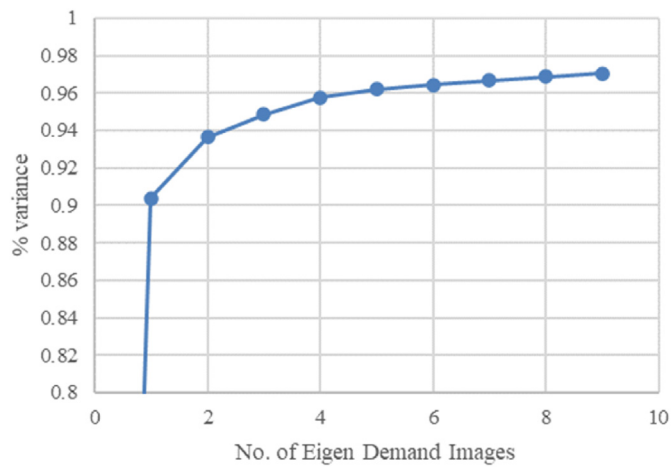

Fig. A1. Results of Elbow method. .

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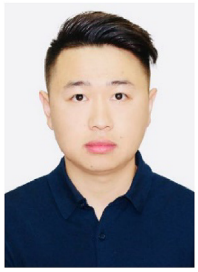

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